EDA for Gender

The GLM Procedure

Class Level Information		
Class	**Levels**	**Values**
Gender	2	Female Male

Number of Observations Read	200
Number of Observations Used	200

EDA for Gender

The GLM Procedure

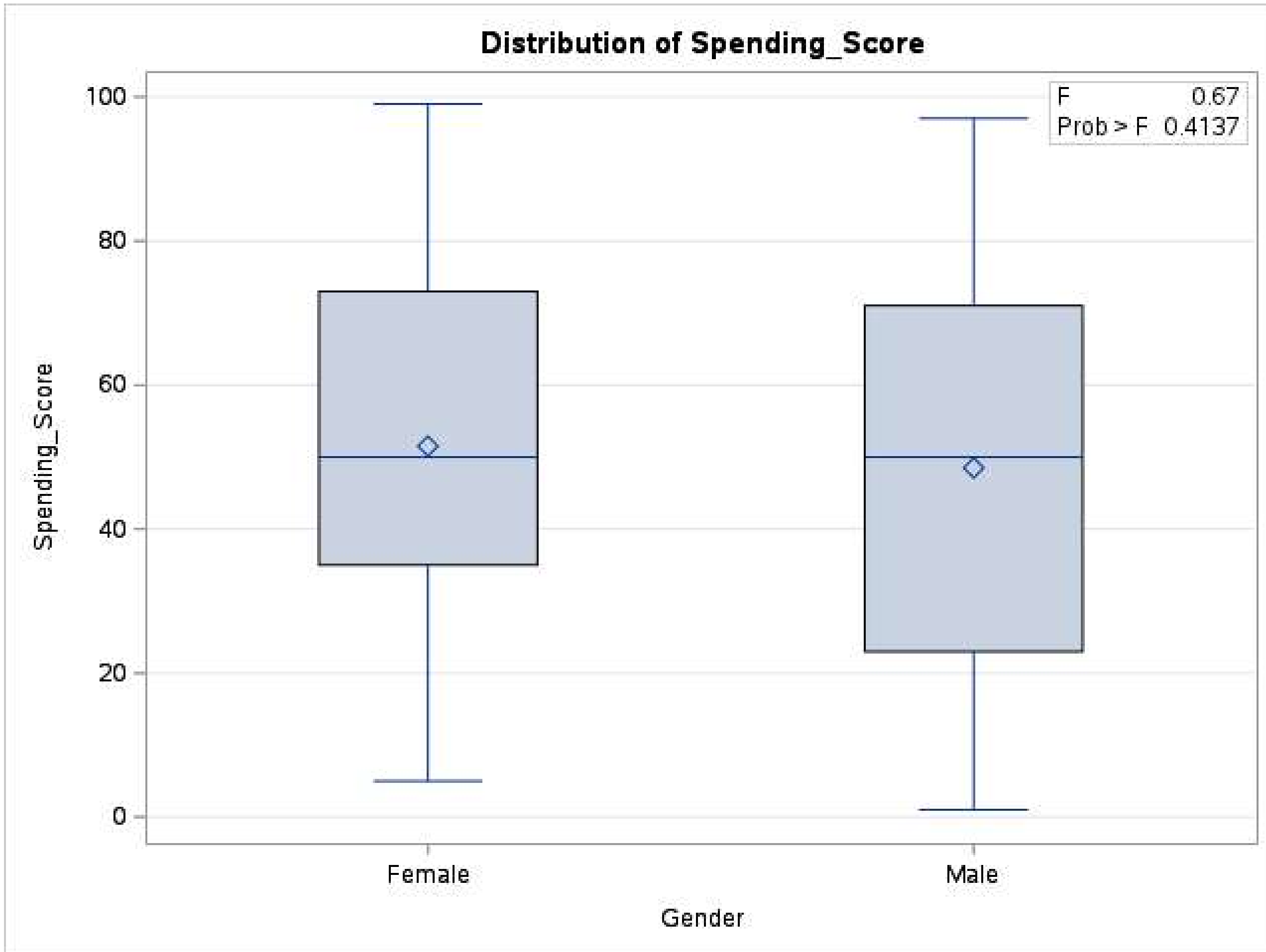
Dependent Variable: Spending_Score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	448.0917	448.0917	0.67	0.4137
Error	198	132255.9083	667.9591		
Corrected Total	199	132704.0000			

R-Square	Coeff Var	Root MSE	Spending_Score Mean
0.003377	51.48388	25.84491	50.20000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Gender	1	448.0917208	448.0917208	0.67	0.4137

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Gender	1	448.0917208	448.0917208	0.67	0.4137



EDA for Gender

The UNIVARIATE Procedure
Variable: Residual

Moments			
N	200	**Sum Weights**	200
Mean	0	**Sum Observations**	0
Std Deviation	25.7798866	**Variance**	664.602554
Skewness	-0.0219163	**Kurtosis**	-0.8450371
Uncorrected SS	132255.908	**Corrected SS**	132255.908

Moments			
Coeff Variation	.	Std Error Mean	1.82291326

Basic Statistical Measures			
Location		Variability	
Mean	0.00000	Std Deviation	25.77989
Median	-0.51907	Variance	664.60255
Mode	-9.52679	Range	96.00000
		Interquartile Range	38.50000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > |t|	1.0000
Sign	M	-2	Pr >= |M|	0.8321
Signed Rank	S	-153	Pr >= |S|	0.8524

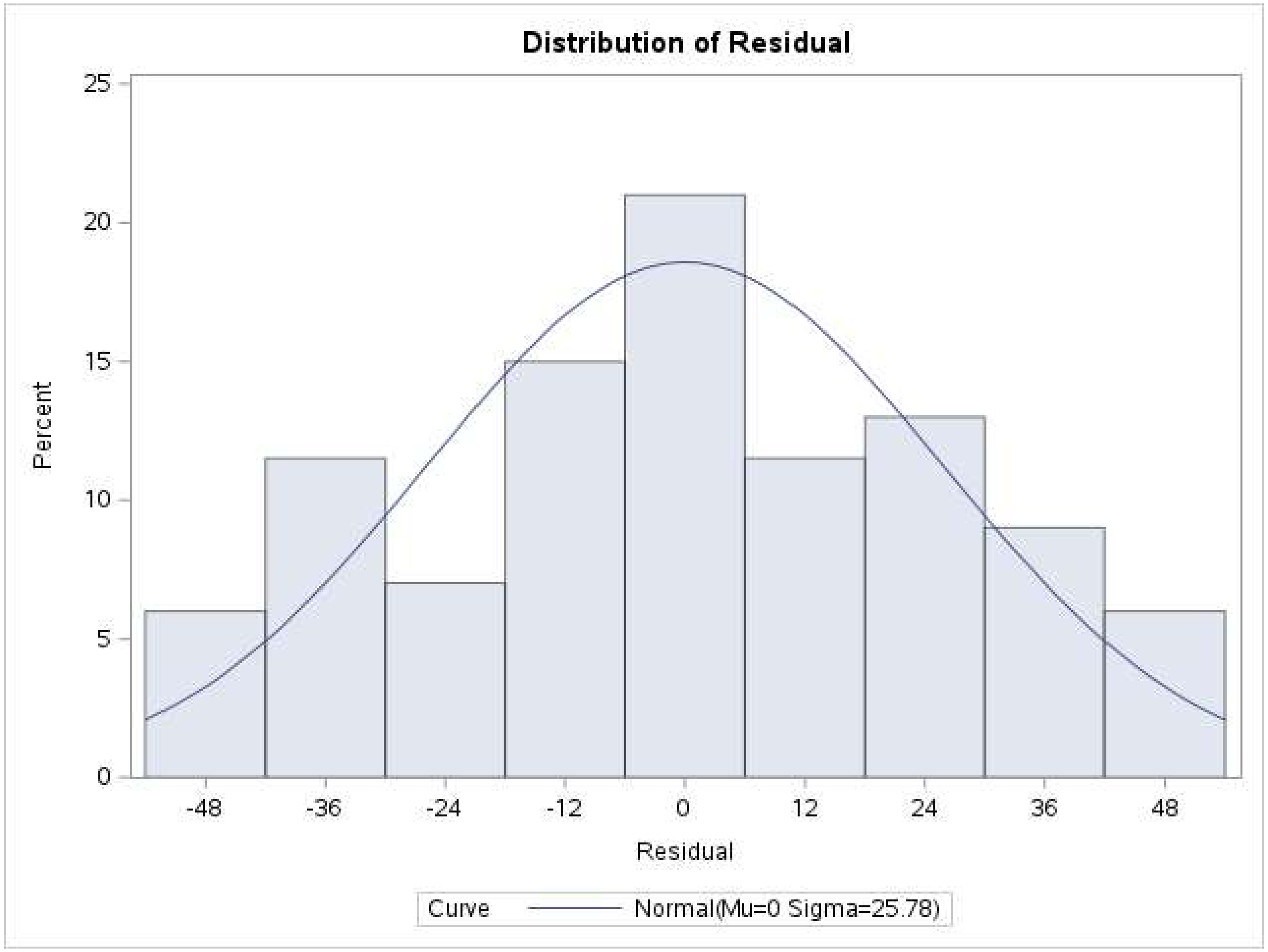
Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.969735	Pr < W	0.0003
Kolmogorov-Smirnov	D	0.063183	Pr > D	0.0493
Cramer-von Mises	W-Sq	0.146177	Pr > W-Sq	0.0267
Anderson-Darling	A-Sq	1.262657	Pr > A-Sq	<0.0050

Quantiles (Definition 5)	
Level	Quantile
100% Max	48.488636
99%	47.980925
95%	42.980925
90%	35.973214
75% Q3	21.473214
50% Median	-0.519075
25% Q1	-17.026786
10%	-37.019075
5%	-44.011364
1%	-47.019075
0% Min	-47.511364

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
-47.5114	159	46.4732	20
-47.5114	157	46.4886	128
-46.5268	141	47.4732	12
-46.5268	23	48.4886	146
-45.5268	7	48.4886	186

EDA for Gender

The UNIVARIATE Procedure



EDA for Gender

The UNIVARIATE Procedure
Fitted Normal Distribution for Residual

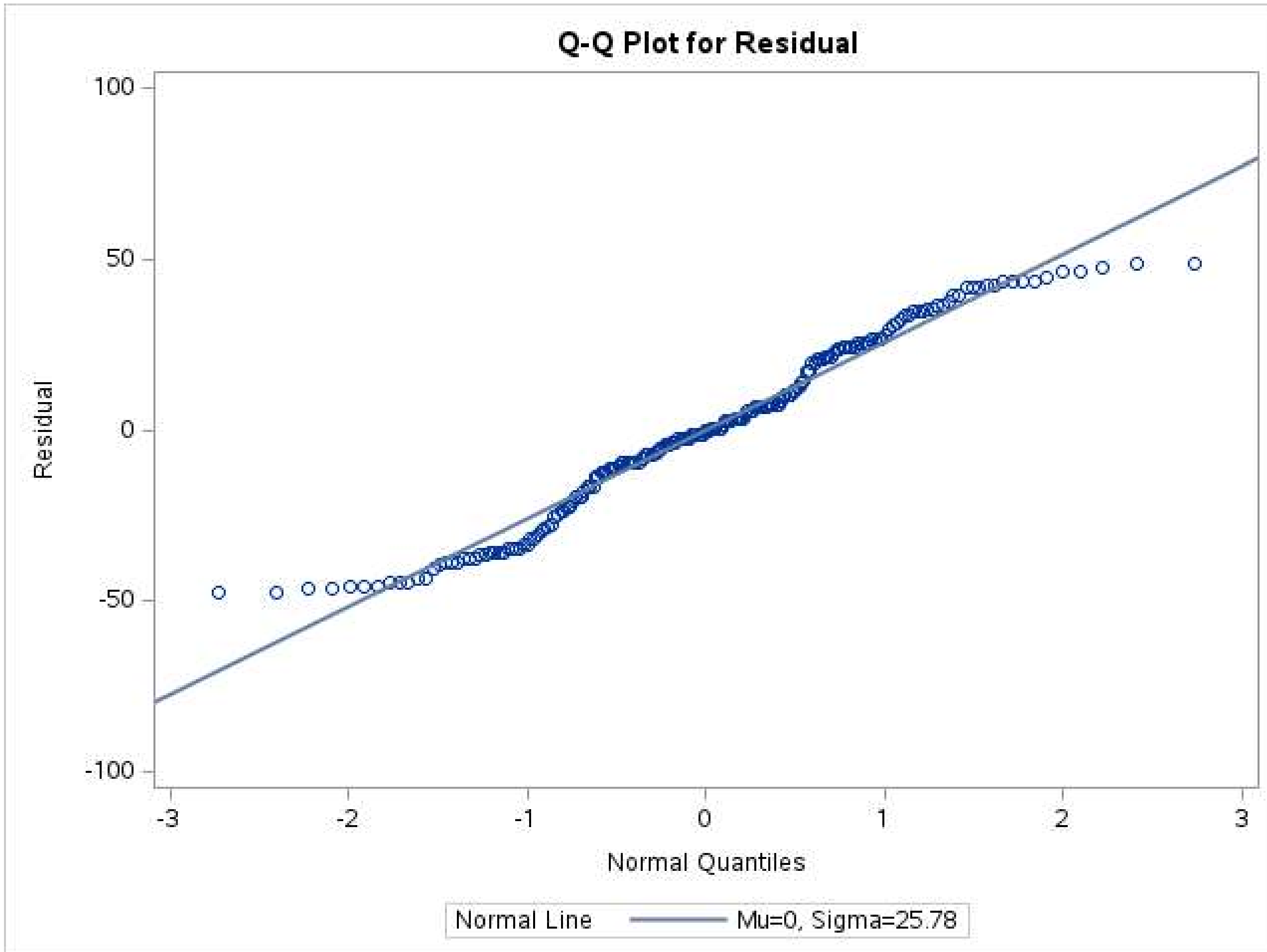
Parameters for Normal Distribution		
Parameter	**Symbol**	**Estimate**
Mean	Mu	0
Std Dev	Sigma	25.77989

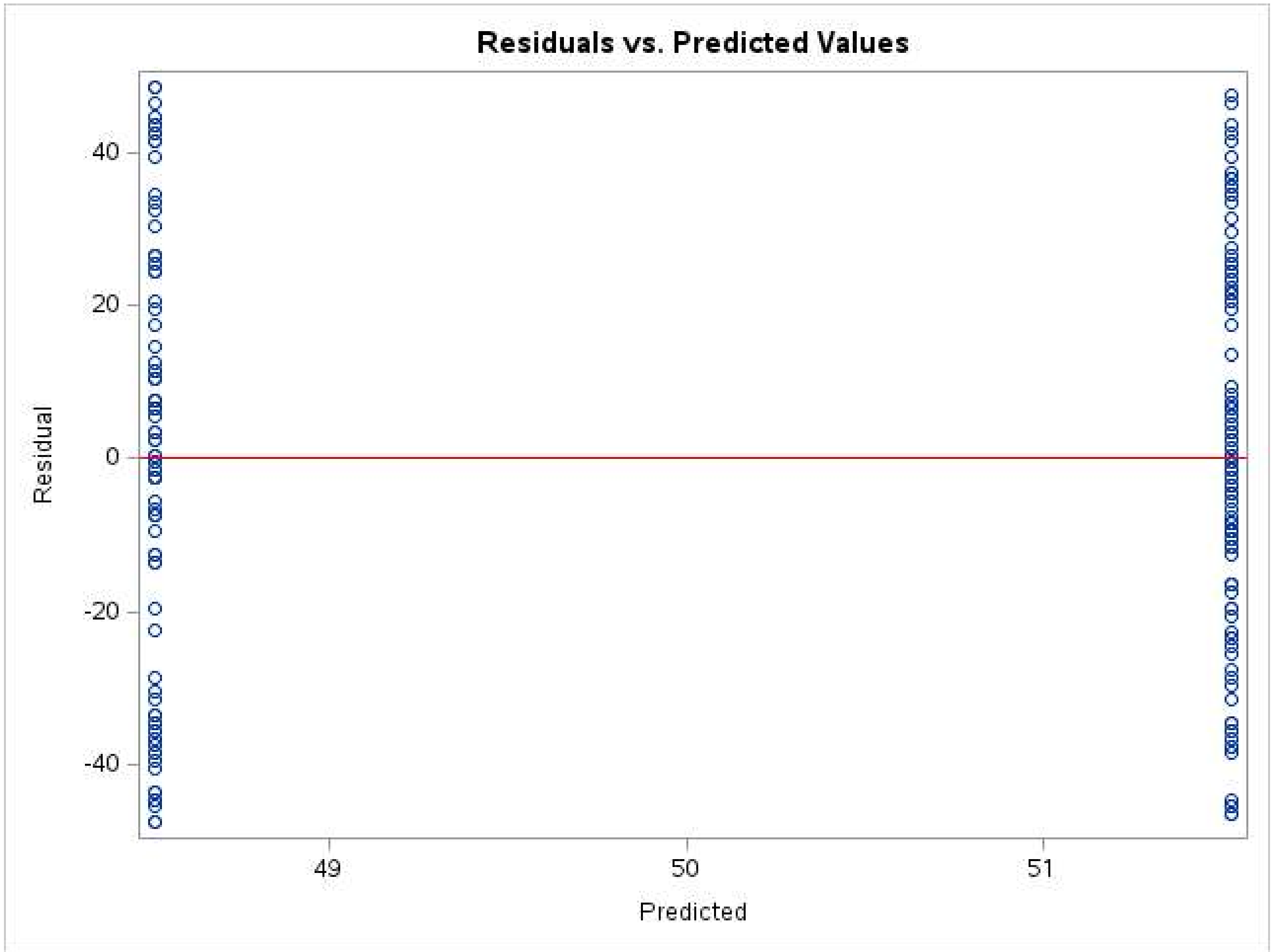
Goodness-of-Fit Tests for Normal Distribution				
Test	**Statistic**		**p Value**	
Kolmogorov-Smirnov	**D**	0.06318297	**Pr > D**	0.049
Cramer-von Mises	**W-Sq**	0.14617712	**Pr > W-Sq**	0.027
Anderson-Darling	**A-Sq**	1.26265682	**Pr > A-Sq**	<0.005

Quantiles for Normal Distribution		
	Quantile	
Percent	**Observed**	**Estimated**
1.0	-47.01907	-59.9730
5.0	-44.01136	-42.4041
10.0	-37.01907	-33.0383
25.0	-17.02679	-17.3883
50.0	-0.51907	0.0000
75.0	21.47321	17.3883
90.0	35.97321	33.0383
95.0	42.98093	42.4041
99.0	47.98093	59.9730

EDA for Gender

The UNIVARIATE Procedure


Q-Q Plot for Residual
100
50
0
-50
-100
Residual
-3
-2
-1
0
1
2
3
Normal Quantiles
Normal Line
Mu=0, Sigma=25.78



Residuals vs. Predicted Values

The GLM Procedure

Class Level Information		
Class	**Levels**	**Values**
Gender	2	Female Male

Number of Observations Read	200
Number of Observations Used	200

Residuals vs. Predicted Values

The GLM Procedure

Dependent Variable: Spending_Score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	448.0917	448.0917	0.67	0.4137
Error	198	132255.9083	667.9591		
Corrected Total	199	132704.0000			

R-Square	Coeff Var	Root MSE	Spending_Score Mean
0.003377	51.48388	25.84491	50.20000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Gender	1	448.0917208	448.0917208	0.67	0.4137

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Gender	1	448.0917208	448.0917208	0.67	0.4137

Residuals vs. Predicted Values

The GLM Procedure

Levene's Test for Homogeneity of Spending_Score Variance ANOVA of Squared Deviations from Group Means					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Gender	1	1836538	1836538	3.70	0.0560
Error	198	98398413	496962		

Residuals vs. Predicted Values

The GLM Procedure

Level of Gender	N	Spending_Score	
		Mean	Std Dev
Female	112	51.5267857	24.1149499
Male	88	48.5113636	27.8967696

Do the spending scores differ across both genders?

The GLM Procedure

Class Level Information		
Class	**Levels**	**Values**
Gender	2	Female Male

Number of Observations Read	200
Number of Observations Used	200

Do the spending scores differ across both genders?

The GLM Procedure

Dependent Variable: Spending_Score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	448.0917	448.0917	0.67	0.4137
Error	198	132255.9083	667.9591		
Corrected Total	199	132704.0000			

R-Square	Coeff Var	Root MSE	Spending_Score Mean
0.003377	51.48388	25.84491	50.20000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Gender	1	448.0917208	448.0917208	0.67	0.4137

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Gender	1	448.0917208	448.0917208	0.67	0.4137

Do the spending scores differ across both genders?

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable Spending_Score Classified by Variable Gender					
Gender	**N**	**Sum of Scores**	**Expected Under H0**	**Std Dev Under H0**	**Mean Score**
Male	88	8613.50	8844.0	406.239673	97.880682
Female	112	11486.50	11256.0	406.239673	102.558036
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr < Z	Pr > |Z|	Pr < Z	Pr > |Z|
8613.500	-0.5662	0.2856	0.5713	0.2860	0.5719
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
0.3219	1	0.5704

EDA for Age_Group

The GLM Procedure

Class Level Information		
Class	**Levels**	**Values**
Age_Group	4	25-39 40-59 60+ Under 25

Number of Observations Read	200
Number of Observations Used	200

EDA for Age_Group

The GLM Procedure

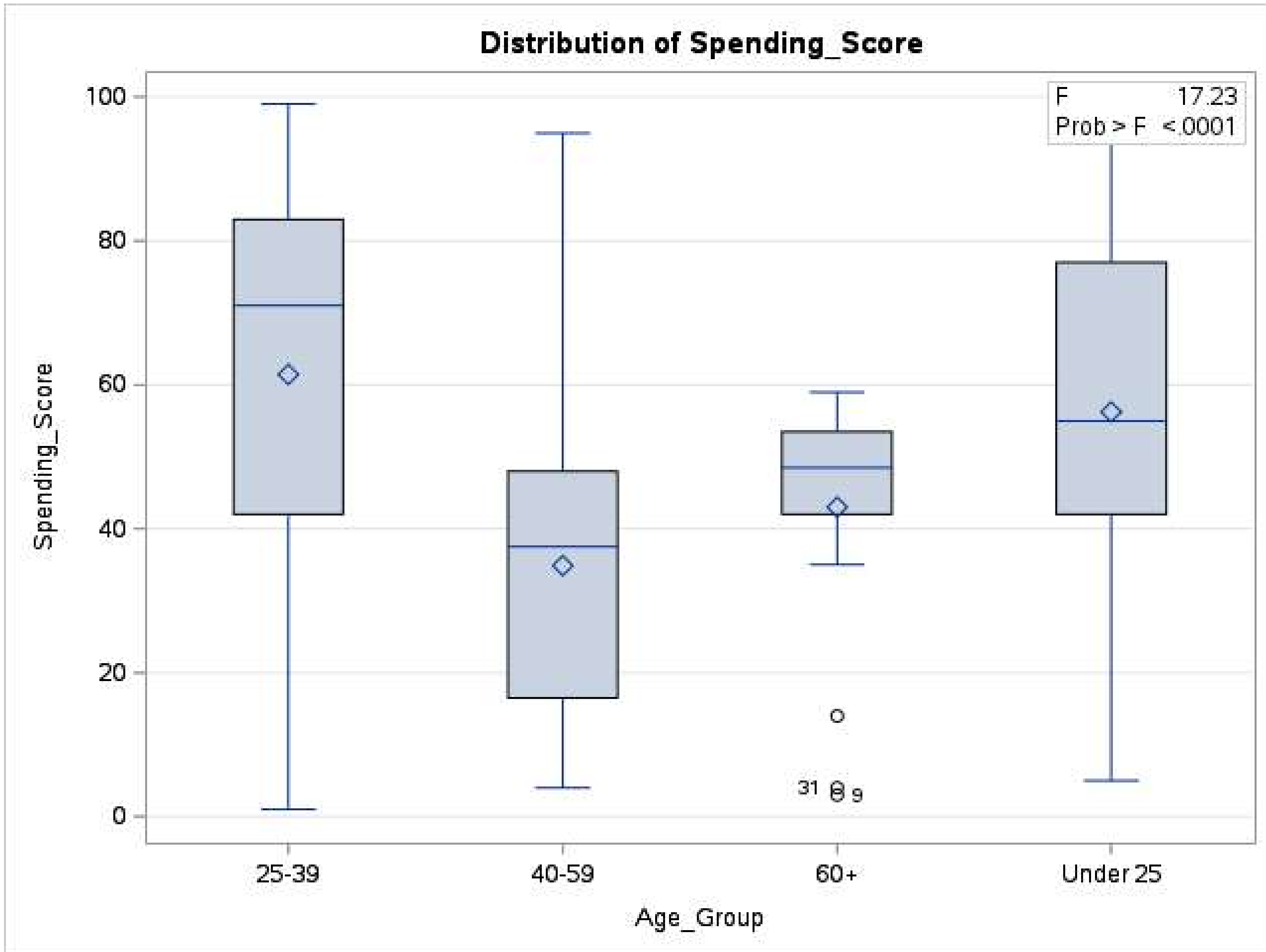
Dependent Variable: Spending_Score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	27691.3577	9230.4526	17.23	<.0001
Error	196	105012.6423	535.7788		
Corrected Total	199	132704.0000			

R-Square	Coeff Var	Root MSE	Spending_Score Mean
0.208670	46.10935	23.14690	50.20000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Age_Group	3	27691.35769	9230.45256	17.23	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Age_Group	3	27691.35769	9230.45256	17.23	<.0001



EDA for Age_Group

The UNIVARIATE Procedure
Variable: Residual

Moments			
N	200	**Sum Weights**	200
Mean	0	**Sum Observations**	0
Std Deviation	22.9717592	**Variance**	527.70172
Skewness	-0.5250979	**Kurtosis**	-0.1133885
Uncorrected SS	105012.642	**Corrected SS**	105012.642

Moments			
Coeff Variation	.	Std Error Mean	1.62434867

Basic Statistical Measures			
Location		Variability	
Mean	0.00000	Std Deviation	22.97176
Median	5.00000	Variance	527.70172
Mode	11.51852	Range	120.62211
		Interquartile Range	33.31752

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > |t|	1.0000
Sign	M	11	Pr >= |M|	0.1374
Signed Rank	S	618.5	Pr >= |S|	0.4518

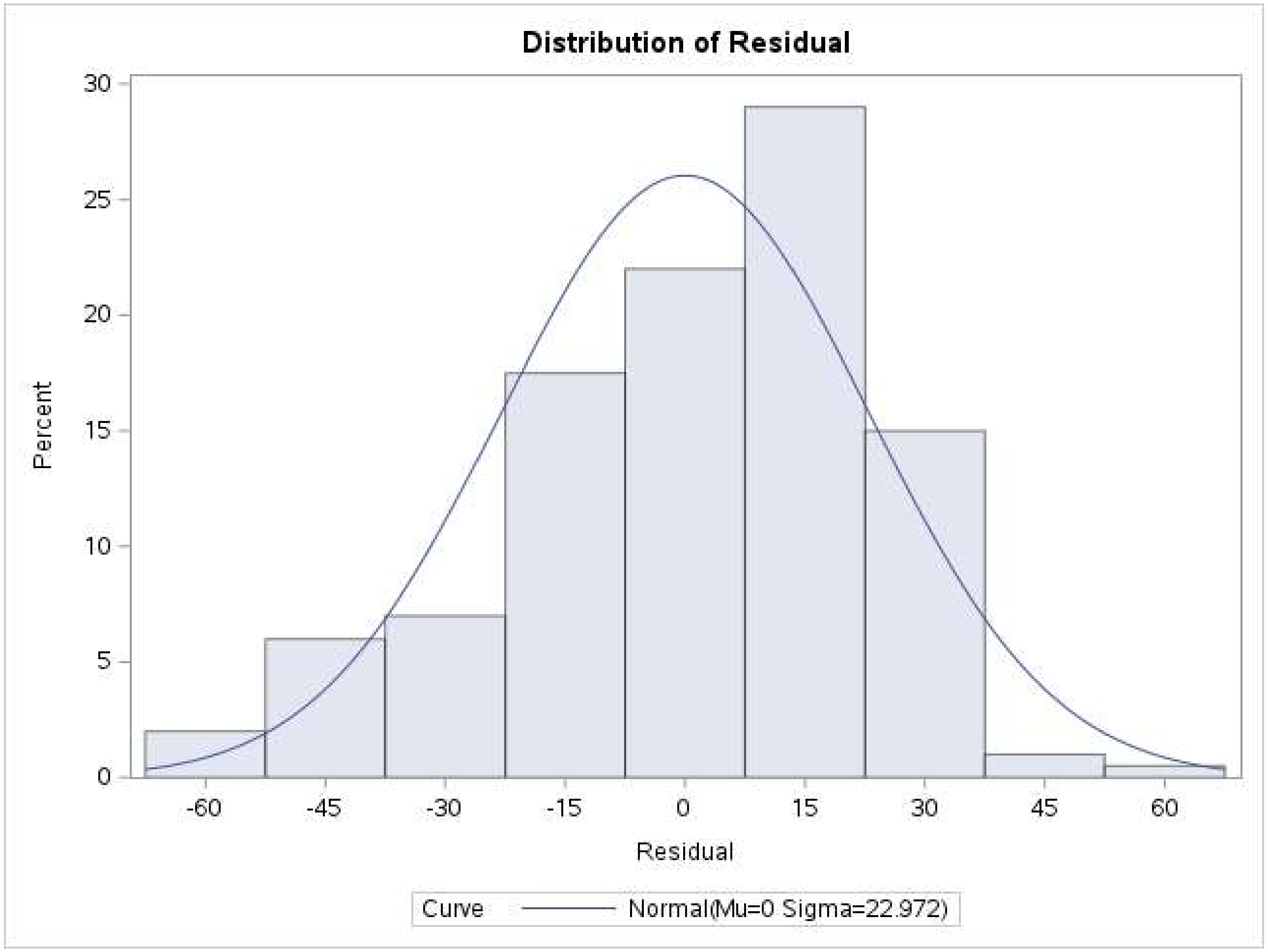
Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.969588	Pr < W	0.0003
Kolmogorov-Smirnov	D	0.094711	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.332455	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	1.952795	Pr > A-Sq	<0.0050

Quantiles (Definition 5)	
Level	Quantile
100% Max	60.1406
99%	37.6307
95%	31.5185
90%	26.5185
75% Q3	15.7593
50% Median	5.0000
25% Q1	-17.5583
10%	-29.8594
5%	-47.3693
1%	-57.9815
0% Min	-60.4815

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
-60.4815	159	35.7429	42
-60.4815	157	36.5185	20
-55.4815	7	37.5185	12
-53.4815	193	37.7429	8
-51.4815	173	60.1406	128

EDA for Age_Group

The UNIVARIATE Procedure



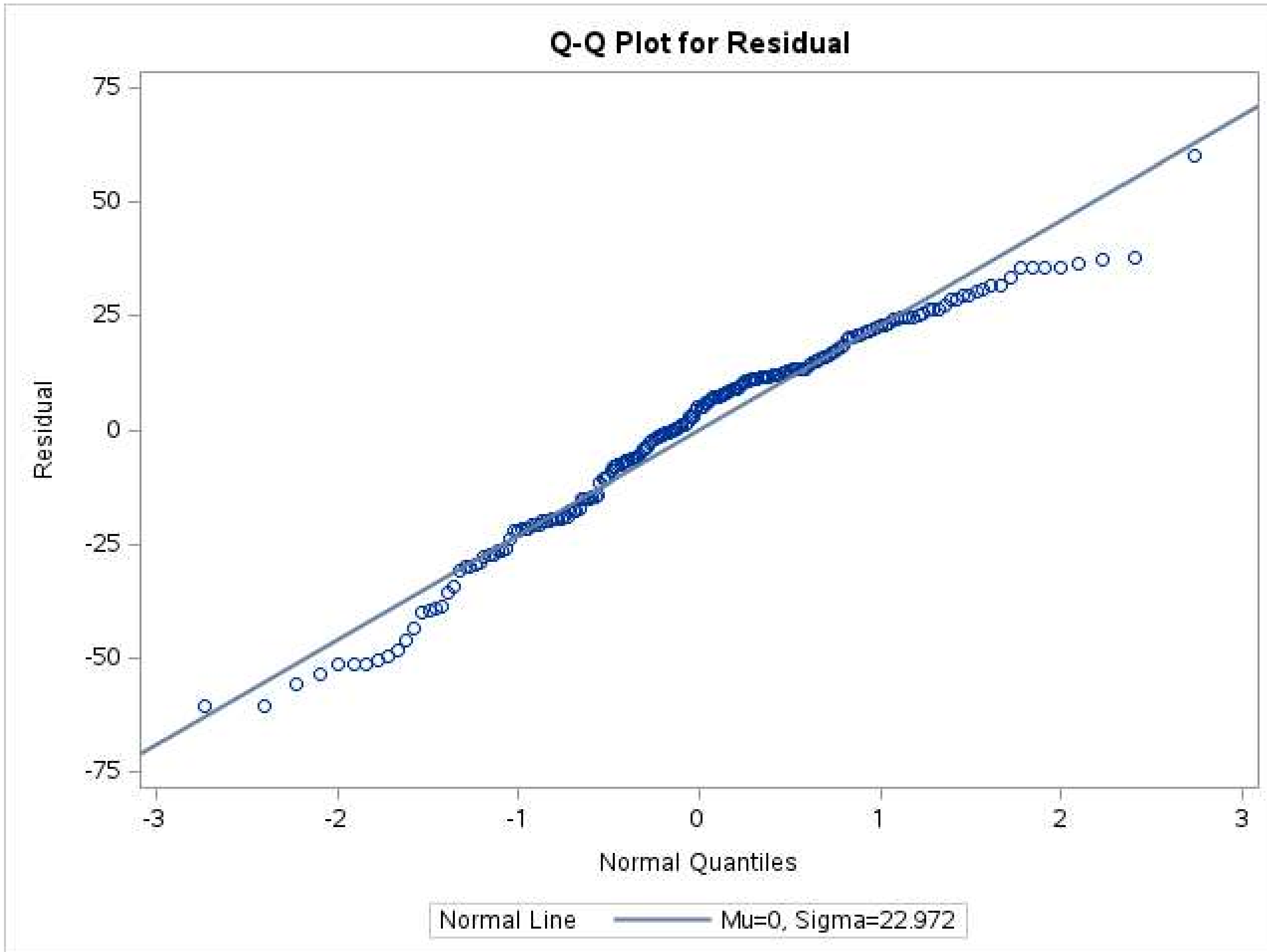
EDA for Age_Group

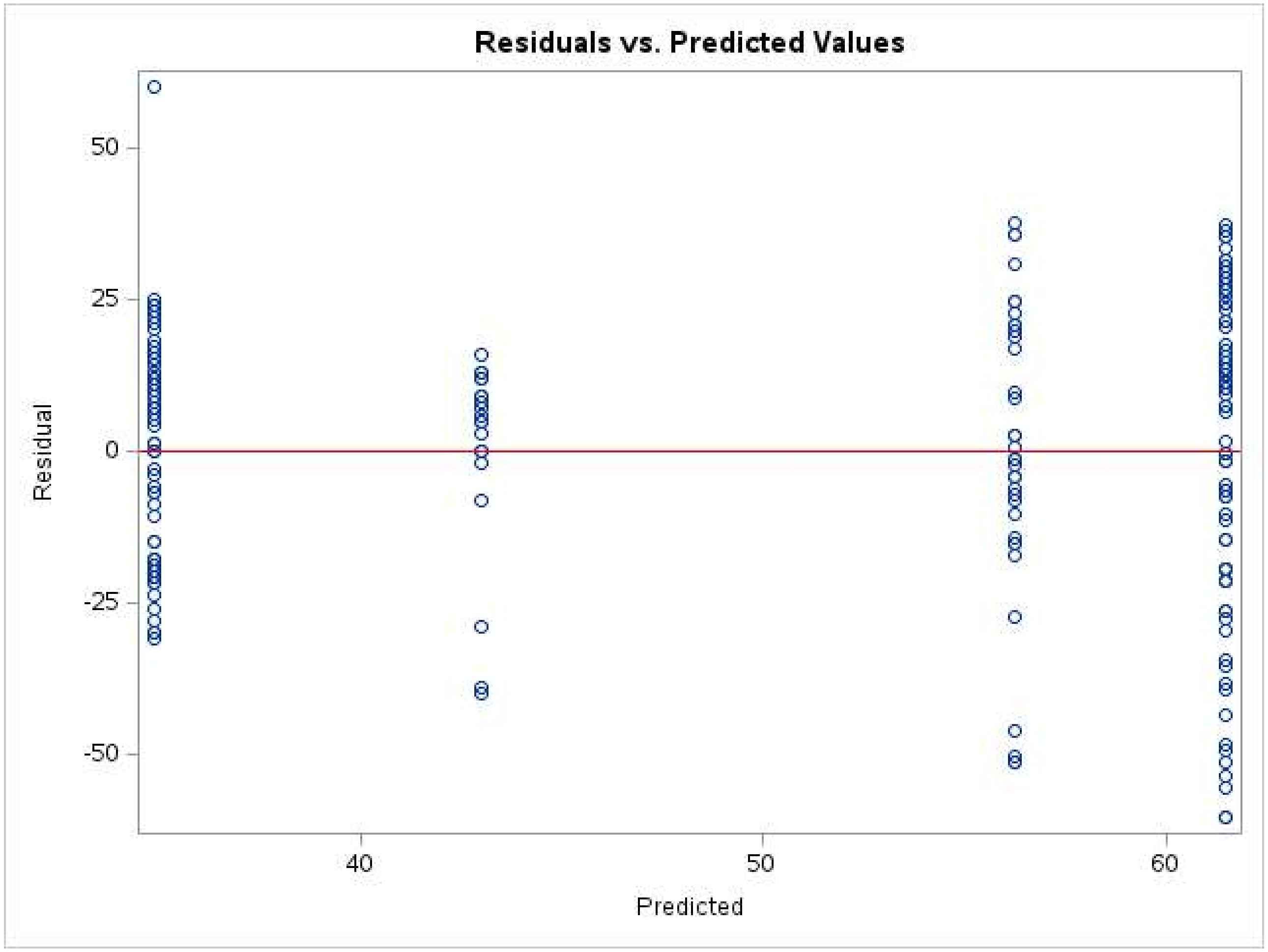
The UNIVARIATE Procedure
Fitted Normal Distribution for Residual

Parameters for Normal Distribution		
Parameter	**Symbol**	**Estimate**
Mean	Mu	0
Std Dev	Sigma	22.97176

Goodness-of-Fit Tests for Normal Distribution				
Test	**Statistic**		**p Value**	
Kolmogorov-Smirnov	**D**	0.09471105	**Pr > D**	<0.010
Cramer-von Mises	**W-Sq**	0.33245460	**Pr > W-Sq**	<0.005
Anderson-Darling	**A-Sq**	1.95279542	**Pr > A-Sq**	<0.005

Quantiles for Normal Distribution		
	Quantile	
Percent	**Observed**	**Estimated**
1.0	-57.98148	-53.4403
5.0	-47.36931	-37.7852
10.0	-29.85938	-29.4395
25.0	-17.55826	-15.4942
50.0	5.00000	0.0000
75.0	15.75926	15.4942
90.0	26.51852	29.4395
95.0	31.51852	37.7852
99.0	37.63069	53.4403


Q-Q Plot for Residual
Residual
75
50
25
0
-25
-50
-75
-3
-2
-1
0
1
2
3
Normal Quantiles
Normal Line
Mu=0, Sigma=22.972



Residuals vs. Predicted Values

The GLM Procedure

Class Level Information		
Class	**Levels**	**Values**
Age_Group	4	25-39 40-59 60+ Under 25

Number of Observations Read	200
Number of Observations Used	200

Residuals vs. Predicted Values

The GLM Procedure

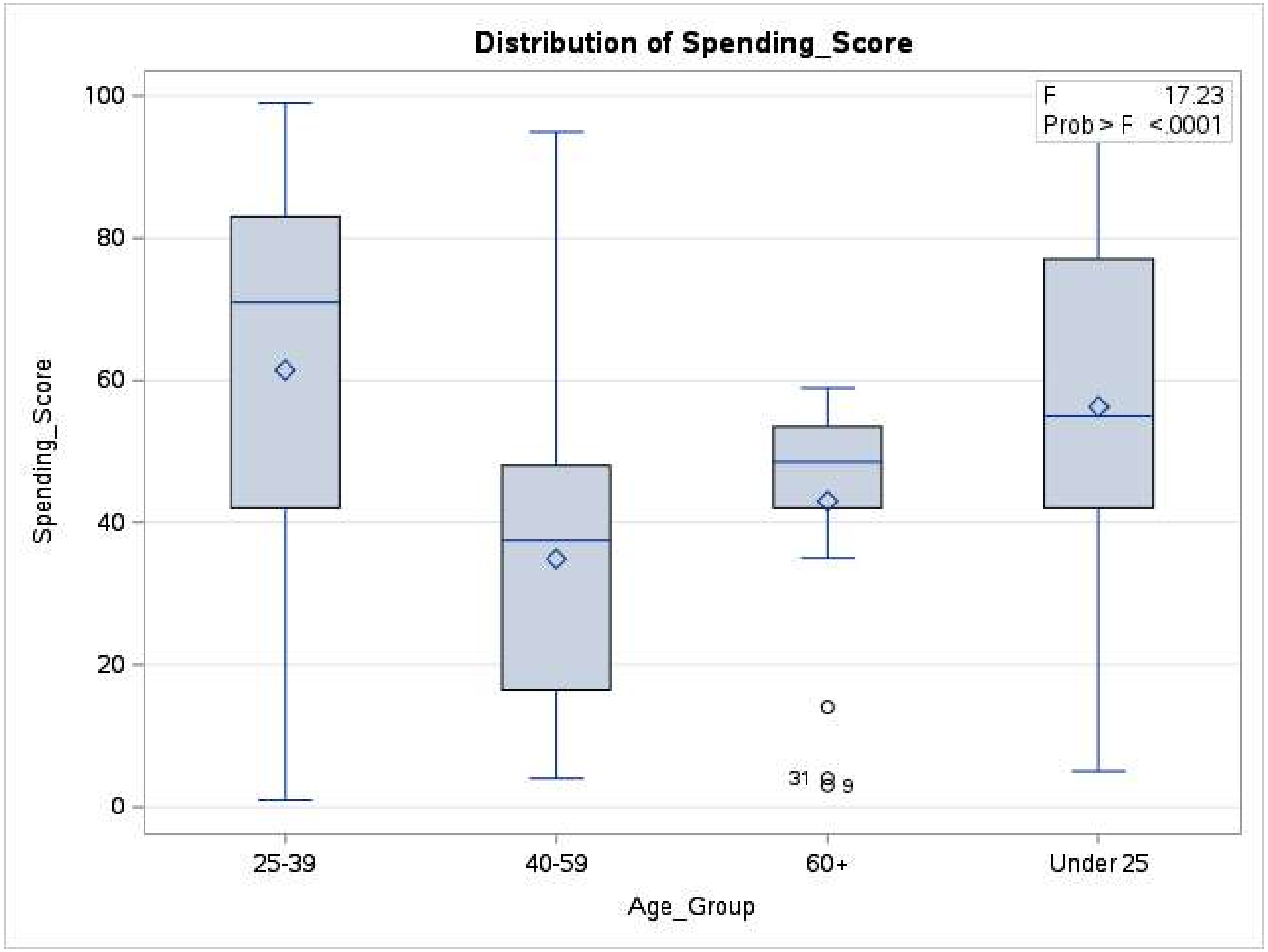
Dependent Variable: Spending_Score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	27691.3577	9230.4526	17.23	<.0001
Error	196	105012.6423	535.7788		
Corrected Total	199	132704.0000			

R-Square	Coeff Var	Root MSE	Spending_Score Mean
0.208670	46.10935	23.14690	50.20000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Age_Group	3	27691.35769	9230.45256	17.23	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Age_Group	3	27691.35769	9230.45256	17.23	<.0001

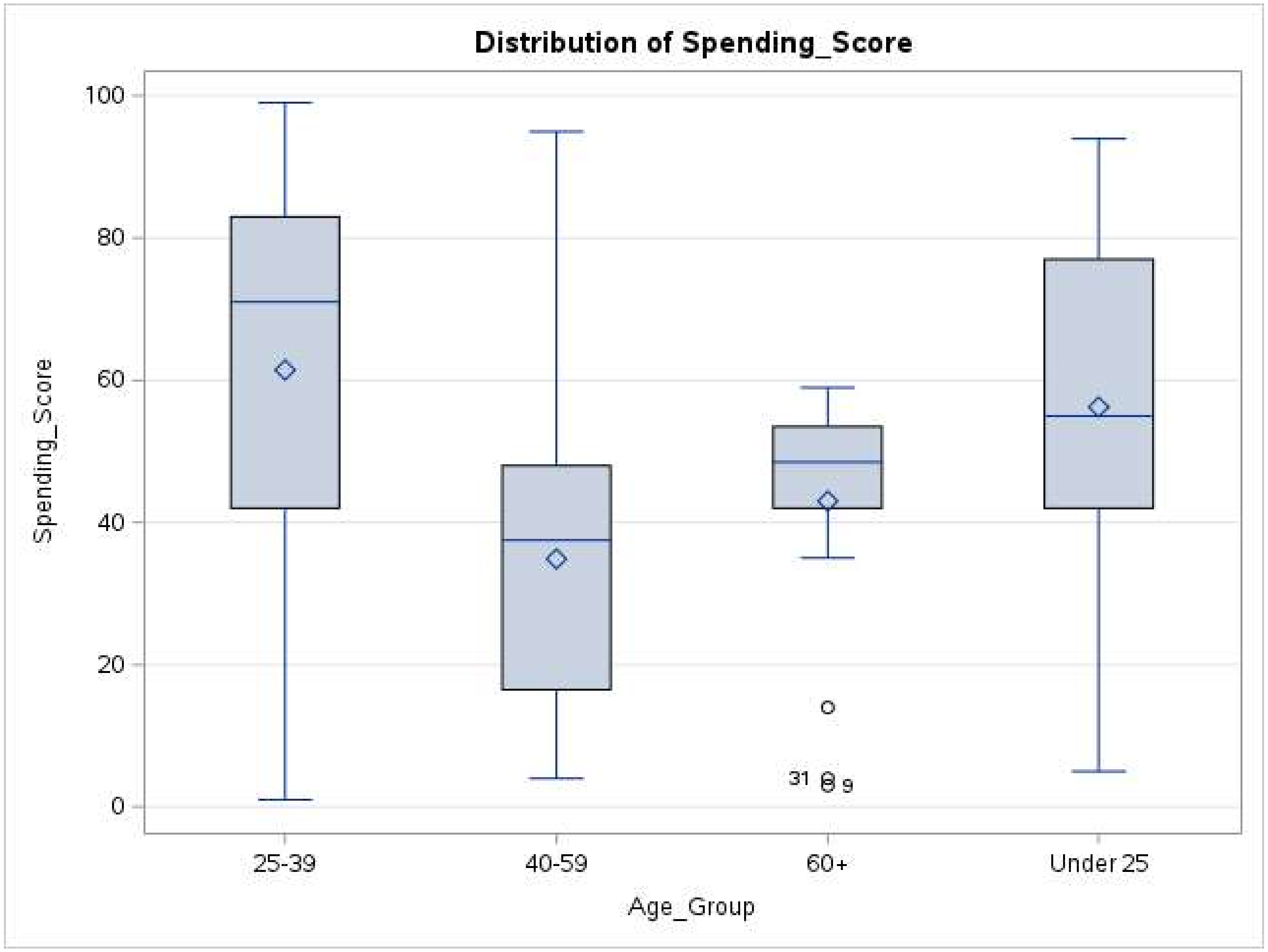


Residuals vs. Predicted Values

The GLM Procedure

Levene's Test for Homogeneity of Spending_Score Variance ANOVA of Squared Deviations from Group Means					
Source	**DF**	**Sum of Squares**	**Mean Square**	**F Value**	**Pr > F**
Age_Group	3	6479307	2159769	4.41	0.0050
Error	196	96054491	490074		

Residuals vs. Predicted Values



Level of Age_Group	N	Spending_Score	
		Mean	Std Dev
25-39	81	61.4814815	26.7881089
40-59	64	34.8593750	18.5368506
60+	20	43.0000000	16.6733320
Under 25	35	56.2571429	24.6592577

Do the annual spendings differ across age groups?

The GLM Procedure

Class Level Information		
Class	Levels	Values
Age_Group	4	25-39 40-59 60+ Under 25

Number of Observations Read	200
Number of Observations Used	200

Do the annual spendings differ across age groups?

The GLM Procedure

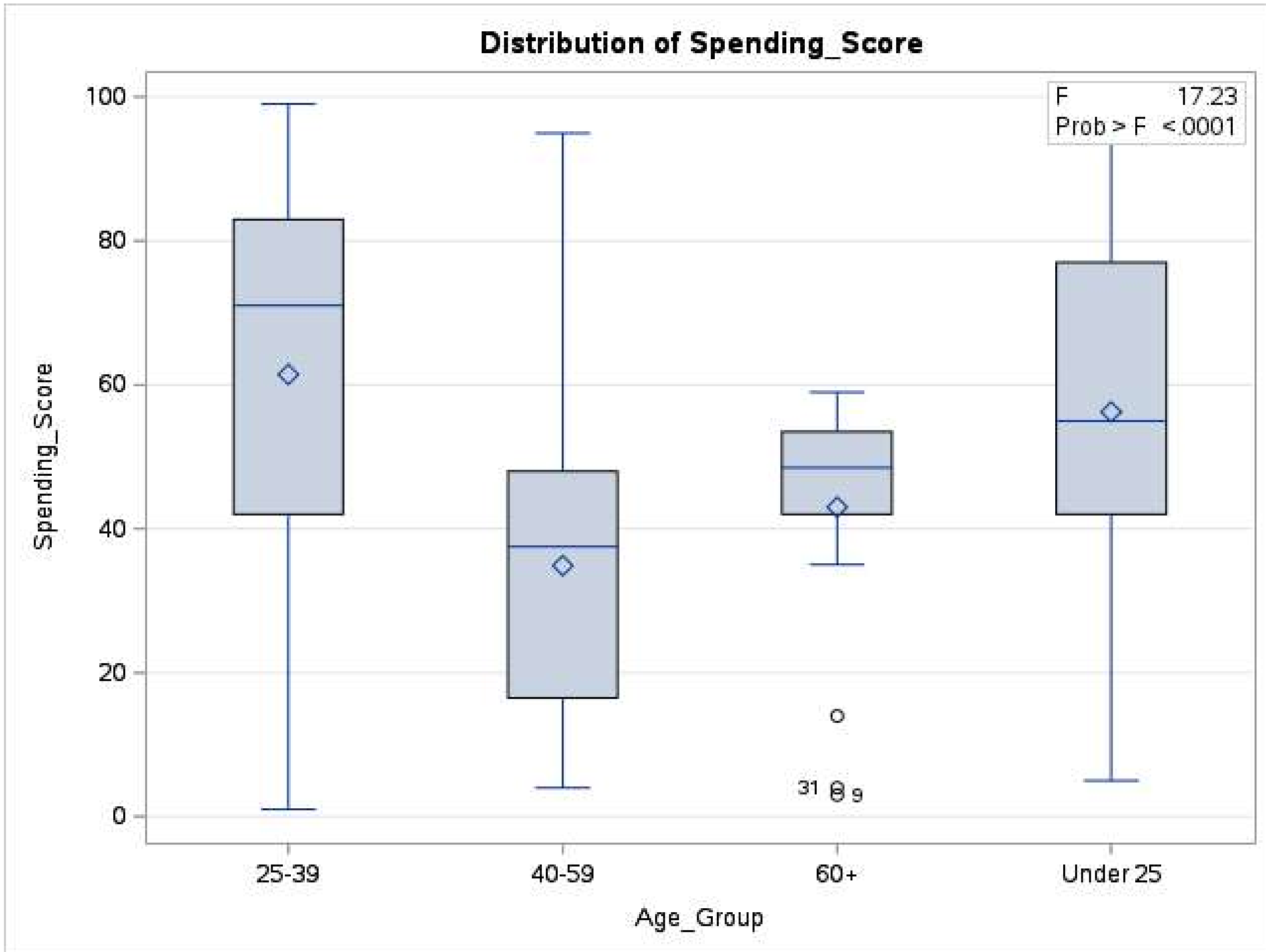
Dependent Variable: Spending_Score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	27691.3577	9230.4526	17.23	<.0001
Error	196	105012.6423	535.7788		
Corrected Total	199	132704.0000			

R-Square	Coeff Var	Root MSE	Spending_Score Mean
0.208670	46.10935	23.14690	50.20000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Age_Group	3	27691.35769	9230.45256	17.23	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Age_Group	3	27691.35769	9230.45256	17.23	<.0001



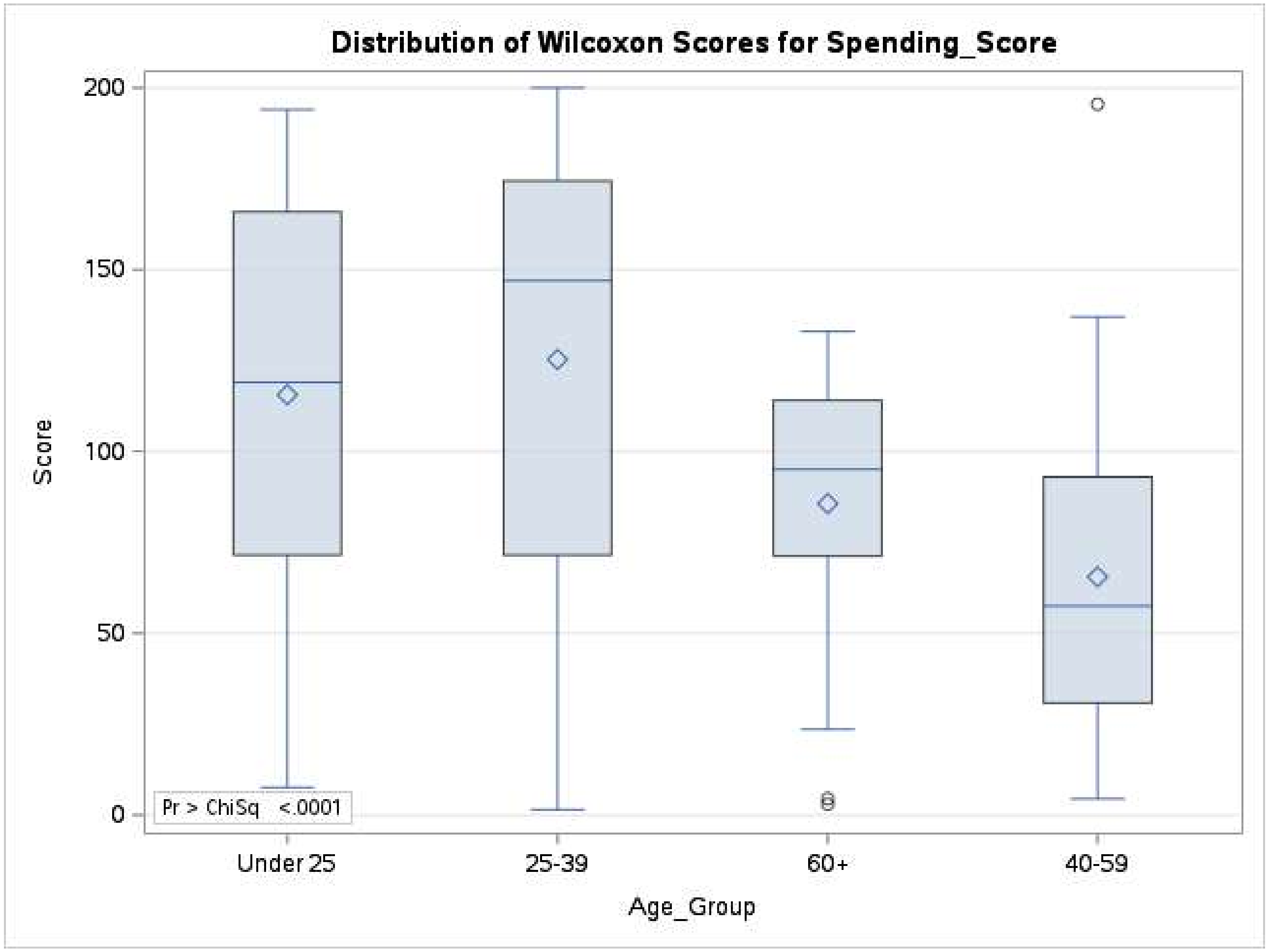
Do the annual spendings differ across age groups?

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable Spending_Score Classified by Variable Age_Group					
Age_Group	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Under 25	35	4044.50	3517.50	310.962472	115.557143
25-39	81	10148.50	8140.50	401.742684	125.290123
60+	20	1714.00	2010.00	245.517943	85.700000
Average scores were used for ties.					

Wilcoxon Scores (Rank Sums) for Variable Spending_Score Classified by Variable Age_Group					
Age_Group	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
40-59	64	4193.00	6432.00	381.760884	65.515625
Average scores were used for ties.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
41.9323	3	<.0001



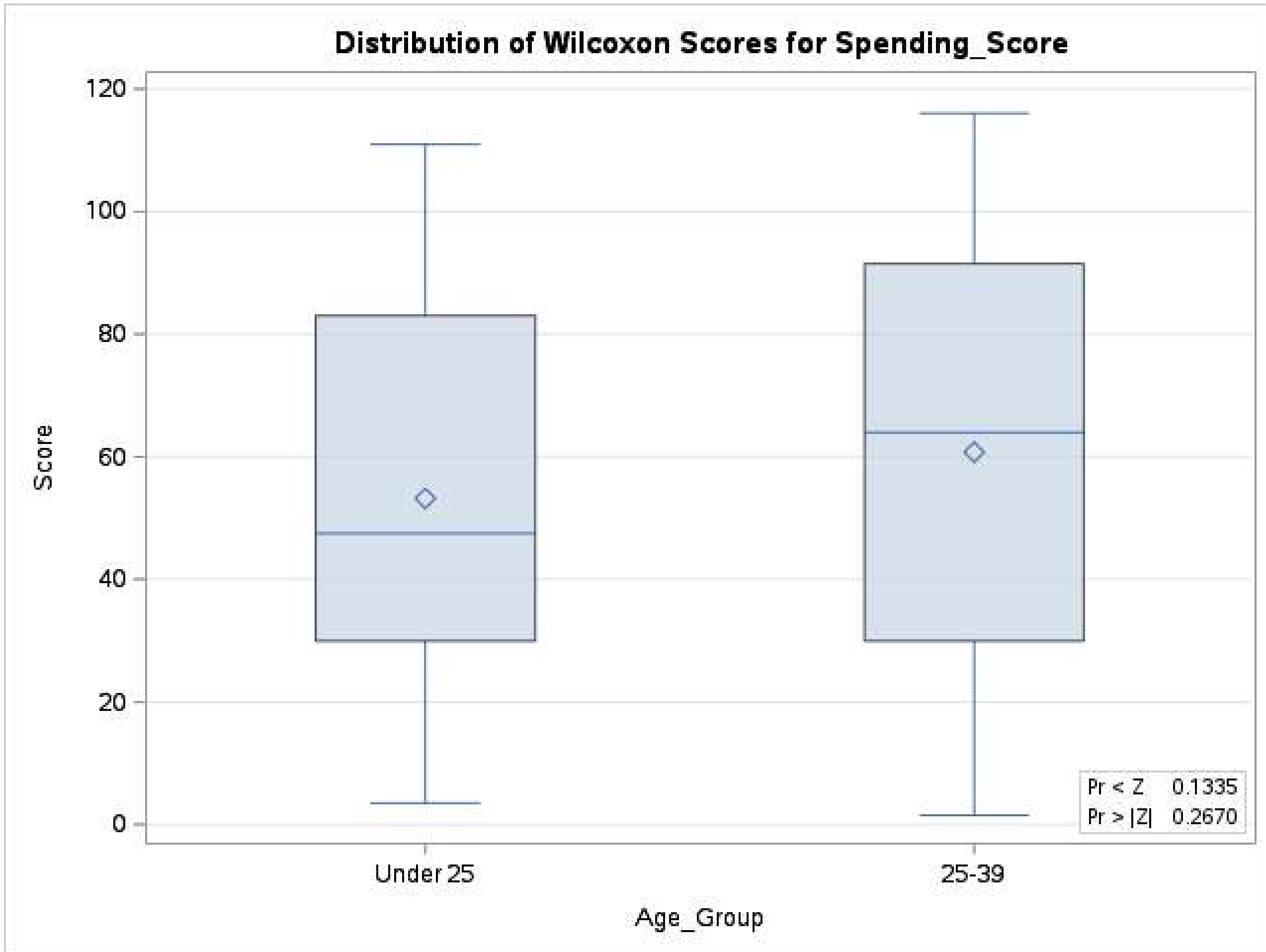
Wilcoxon: Under 25 vs 25-39

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable Spending_Score Classified by Variable Age_Group					
Age_Group	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Under 25	35	1862.50	2047.50	166.215030	53.214286
25-39	81	4923.50	4738.50	166.215030	60.783951
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr < Z	Pr > |Z|	Pr < Z	Pr > |Z|
1862.500	-1.1100	0.1335	0.2670	0.1347	0.2693
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
1.2388	1	0.2657



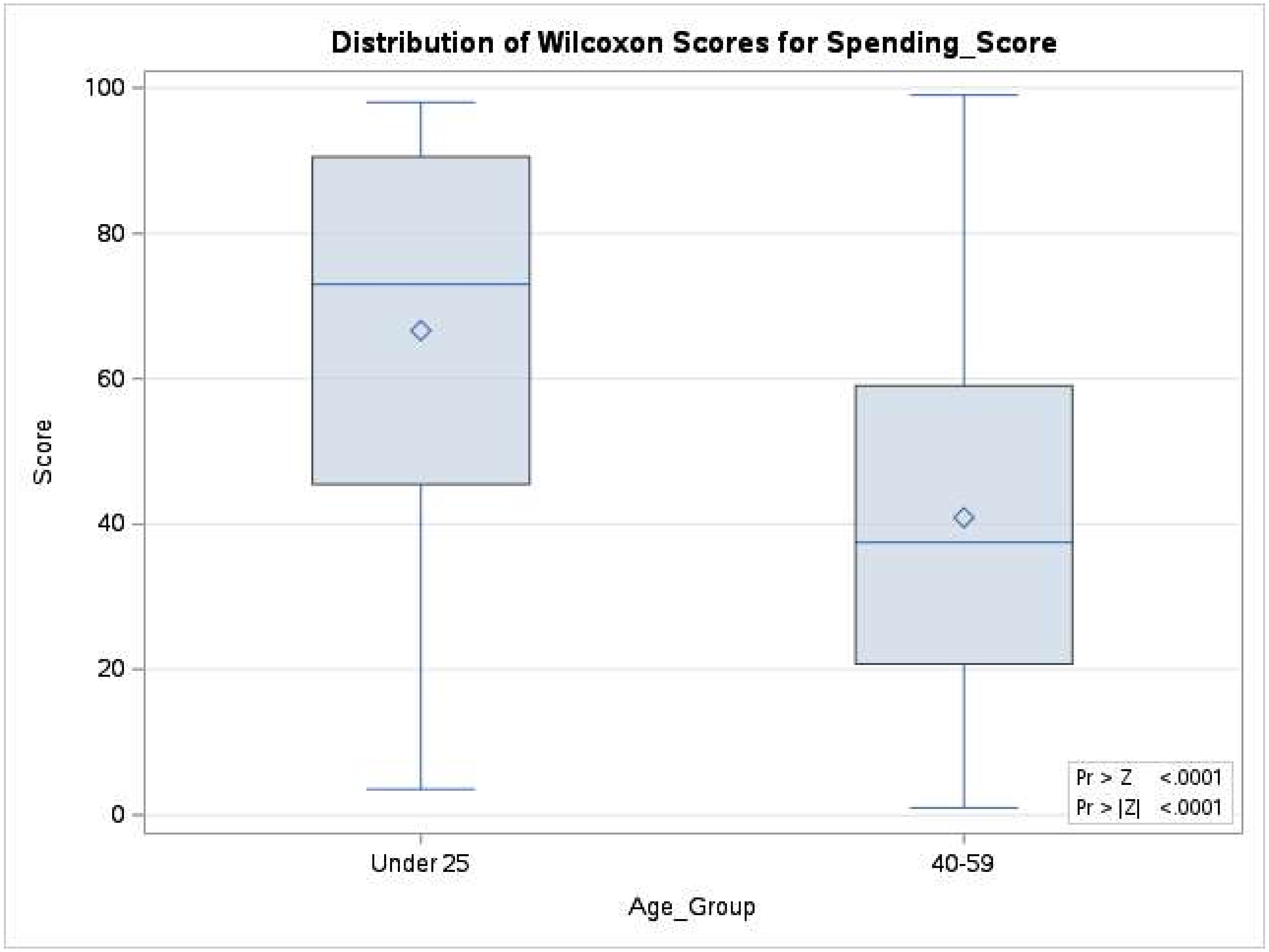
Wilcoxon: Under 25 vs 40-59

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable Spending_Score Classified by Variable Age_Group					
Age_Group	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Under 25	35	2331.50	1750.0	136.585025	66.614286
40-59	64	2618.50	3200.0	136.585025	40.914063
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr > Z	Pr > |Z|	Pr > Z	Pr > |Z|
2331.500	4.2538	<.0001	<.0001	<.0001	<.0001
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
18.1256	1	<.0001



Wilcoxon: Under 25 vs 60+

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable Spending_Score Classified by Variable Age_Group

Age_Group	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
Under 25	35	1110.50	980.0	57.112477	31.728571
60+	20	429.50	560.0	57.112477	21.475000

Average scores were used for ties.

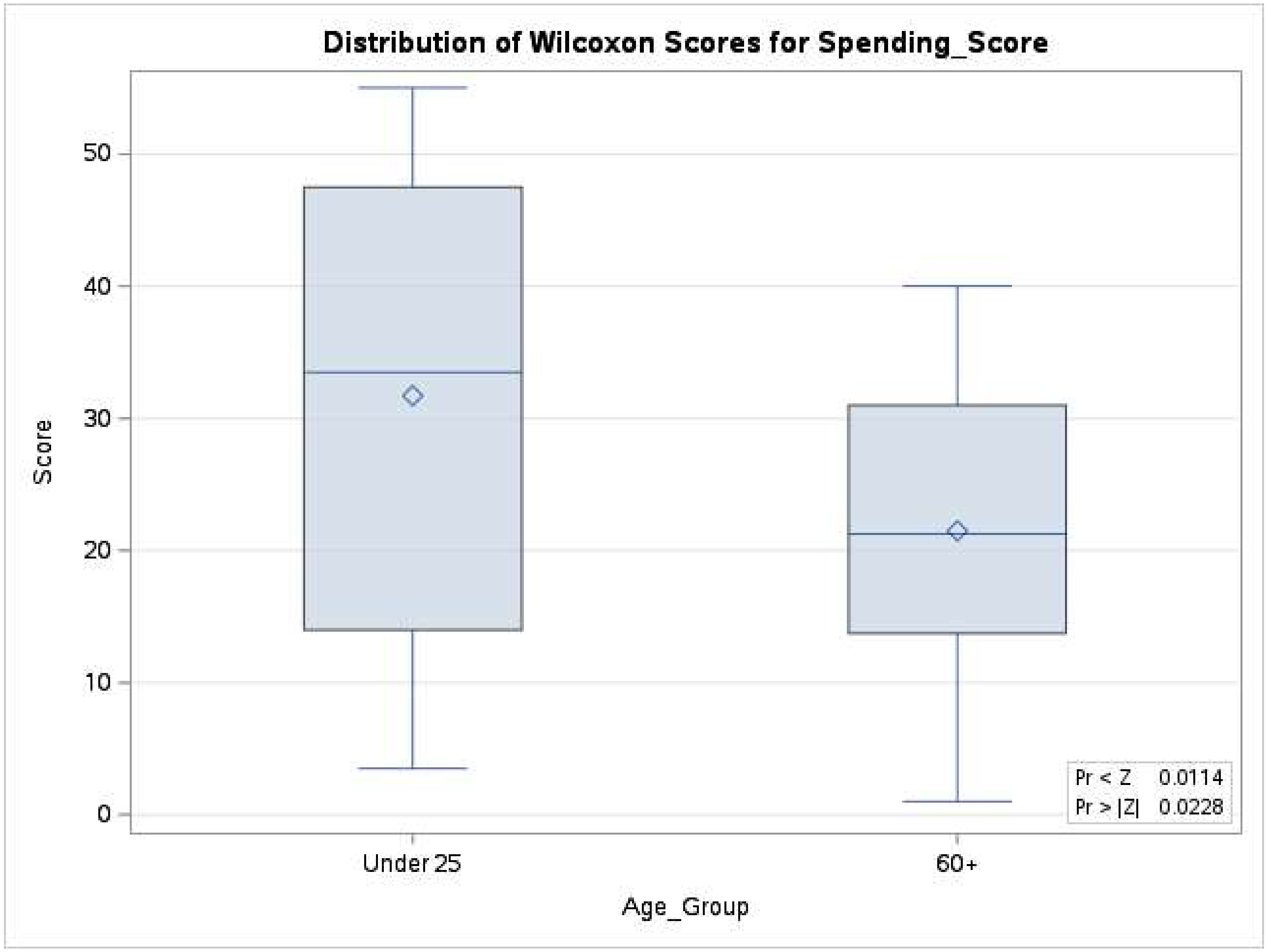
Wilcoxon Two-Sample Test

				t Approximation	
Statistic	Z	Pr < Z	Pr > |Z|	Pr < Z	Pr > |Z|
429.5000	-2.2762	0.0114	0.0228	0.0134	0.0268

Z includes a continuity correction of 0.5.

Kruskal-Wallis Test

Chi-Square	DF	Pr > ChiSq
5.2211	1	0.0223



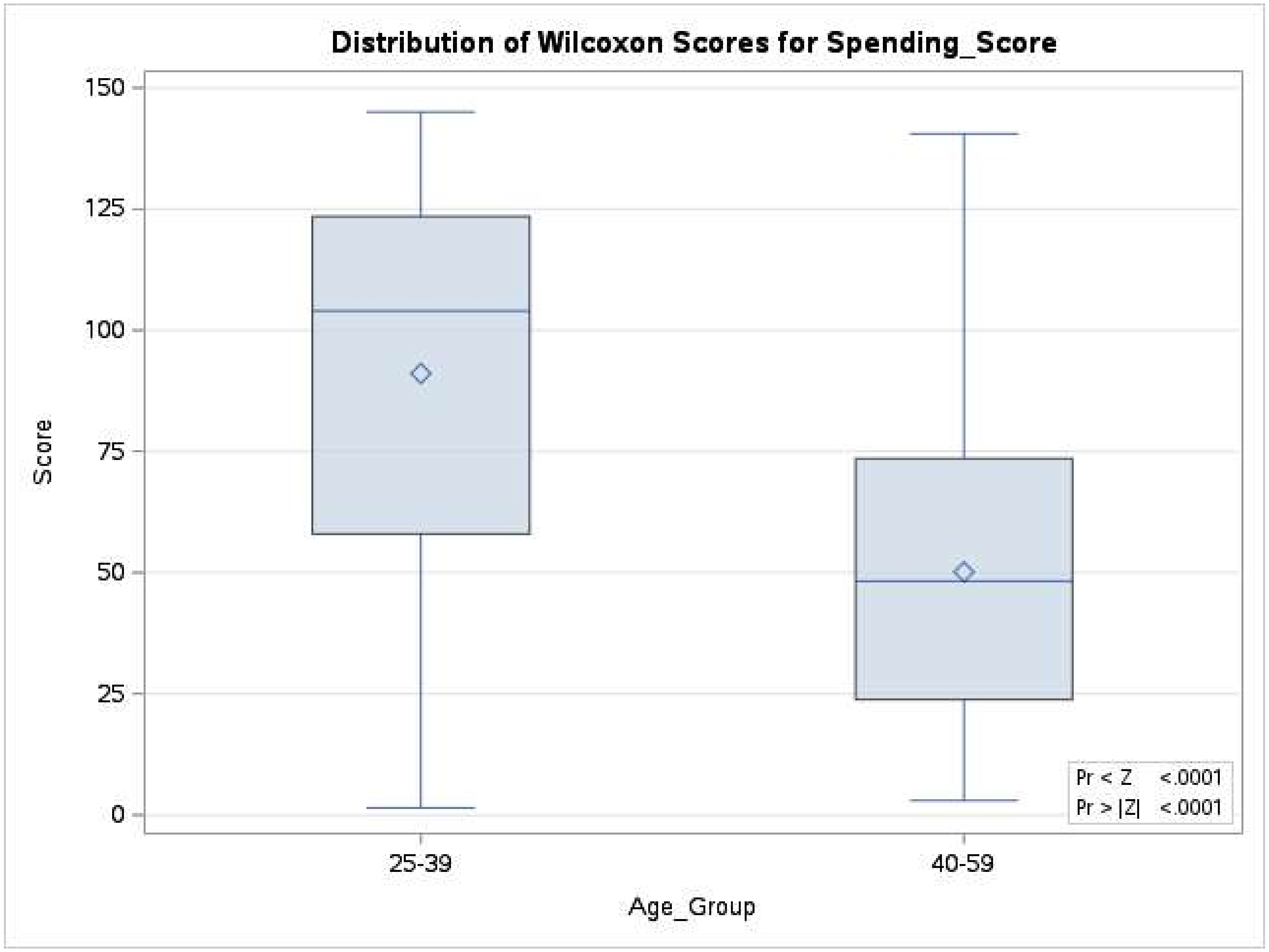
Wilcoxon: 25-39 vs 40-59

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable Spending_Score Classified by Variable Age_Group					
Age_Group	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
25-39	81	7376.0	5913.0	251.096162	91.061728
40-59	64	3209.0	4672.0	251.096162	50.140625
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr < Z	Pr > |Z|	Pr < Z	Pr > |Z|
3209.000	-5.8245	<.0001	<.0001	<.0001	<.0001
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
33.9476	1	<.0001



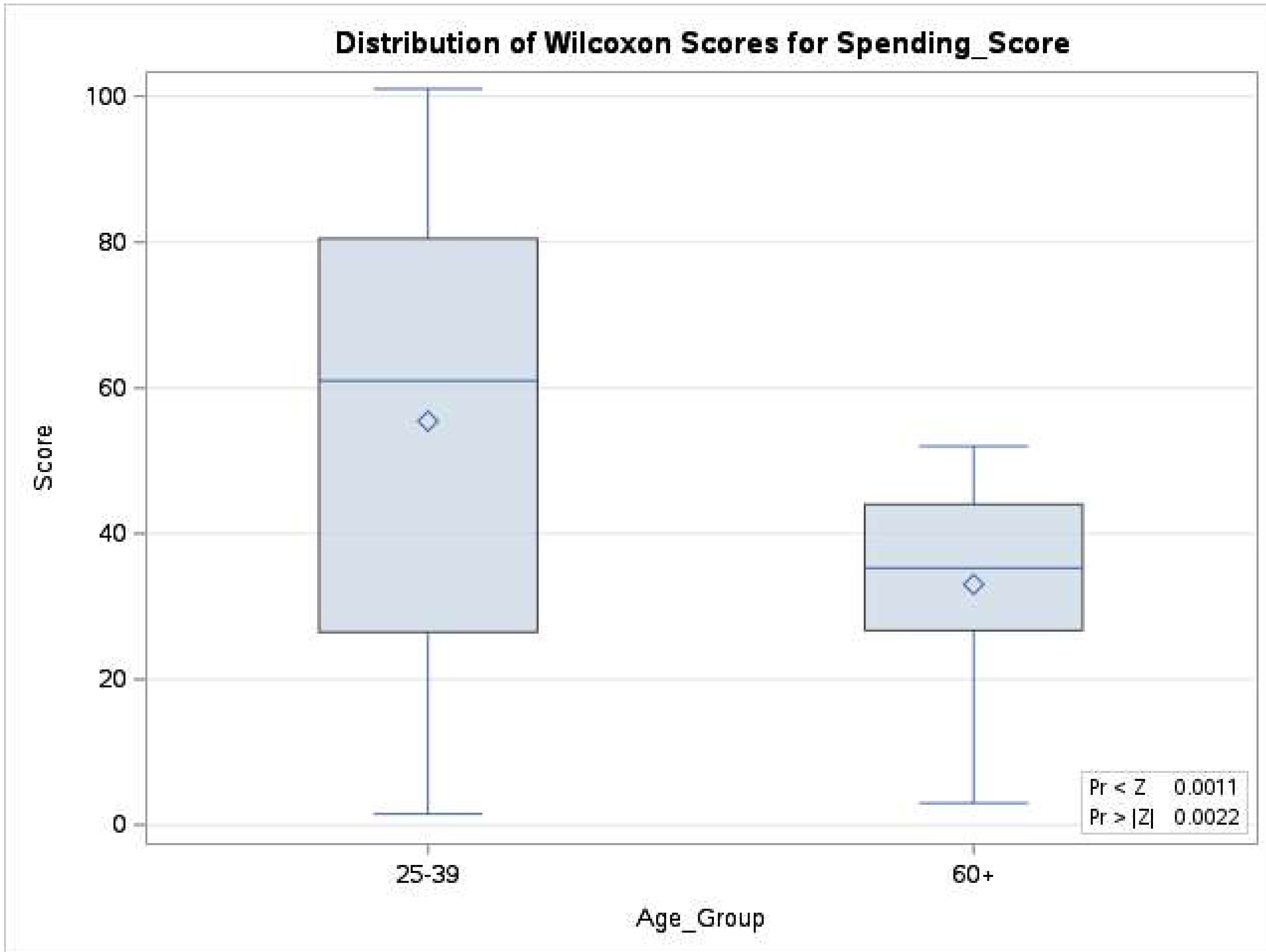
Wilcoxon: 25-39 vs 60+

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable Spending_Score Classified by Variable Age_Group					
Age_Group	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
25-39	81	4491.0	4131.0	117.316594	55.444444
60+	20	660.0	1020.0	117.316594	33.000000
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr < Z	Pr > |Z|	Pr < Z	Pr > |Z|
660.0000	-3.0644	0.0011	0.0022	0.0014	0.0028
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
9.4164	1	0.0022



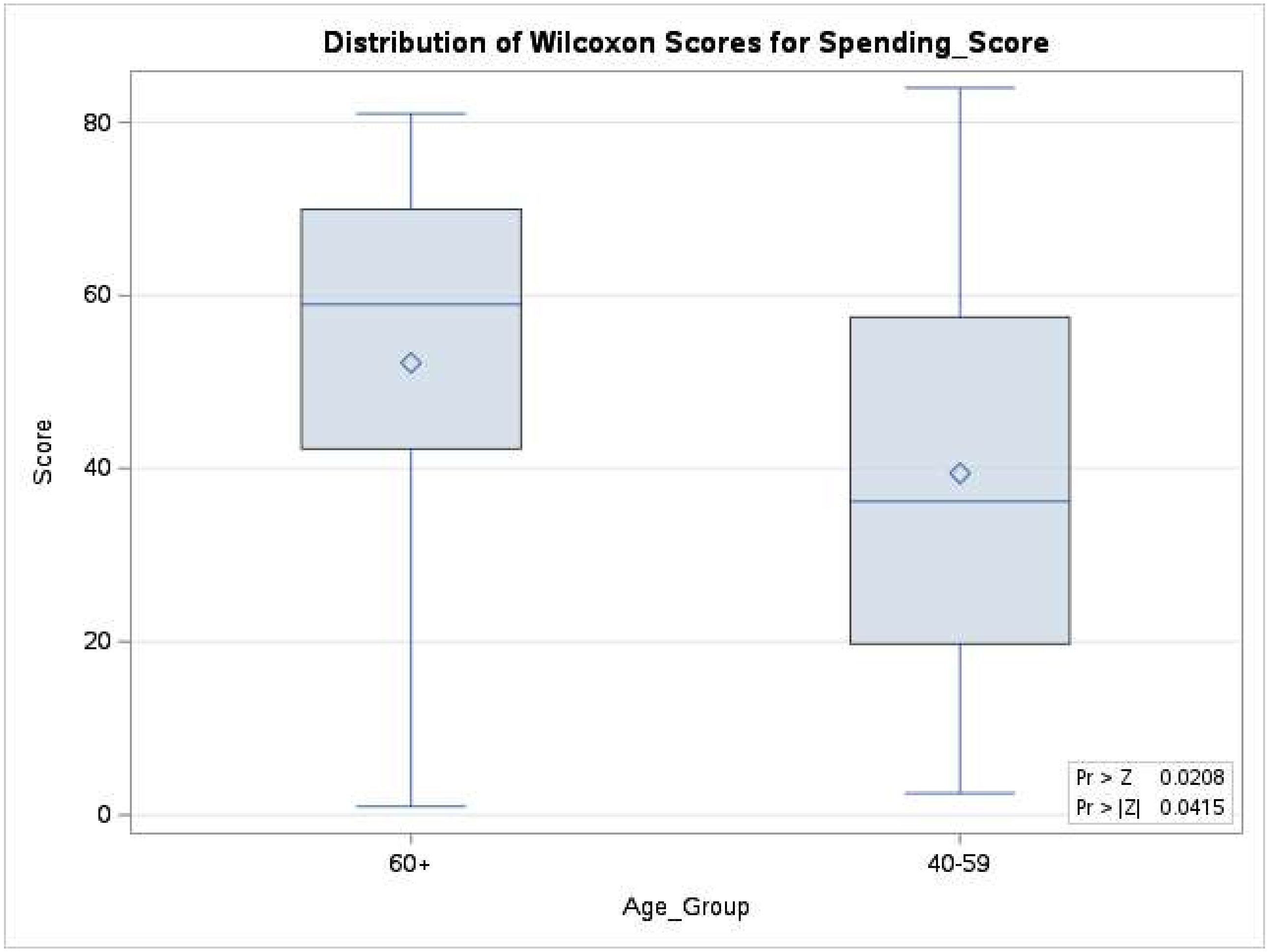
Wilcoxon: 40-59 vs 60+

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable Spending_Score Classified by Variable Age_Group					
Age_Group	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
60+	20	1044.50	850.0	95.174689	52.225000
40-59	64	2525.50	2720.0	95.174689	39.460938
Average scores were used for ties.					

Wilcoxon Two-Sample Test					
				t Approximation	
Statistic	Z	Pr > Z	Pr > |Z|	Pr > Z	Pr > |Z|
1044.500	2.0384	0.0208	0.0415	0.0223	0.0447
Z includes a continuity correction of 0.5.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
4.1763	1	0.0410



Tukey HSD Post-Hoc for Age_Group ANOVA

The GLM Procedure

Class Level Information		
Class	Levels	Values
Age_Group	4	25-39 40-59 60+ Under 25

Number of Observations Read	200
Number of Observations Used	200

Tukey HSD Post-Hoc for Age_Group ANOVA

The GLM Procedure

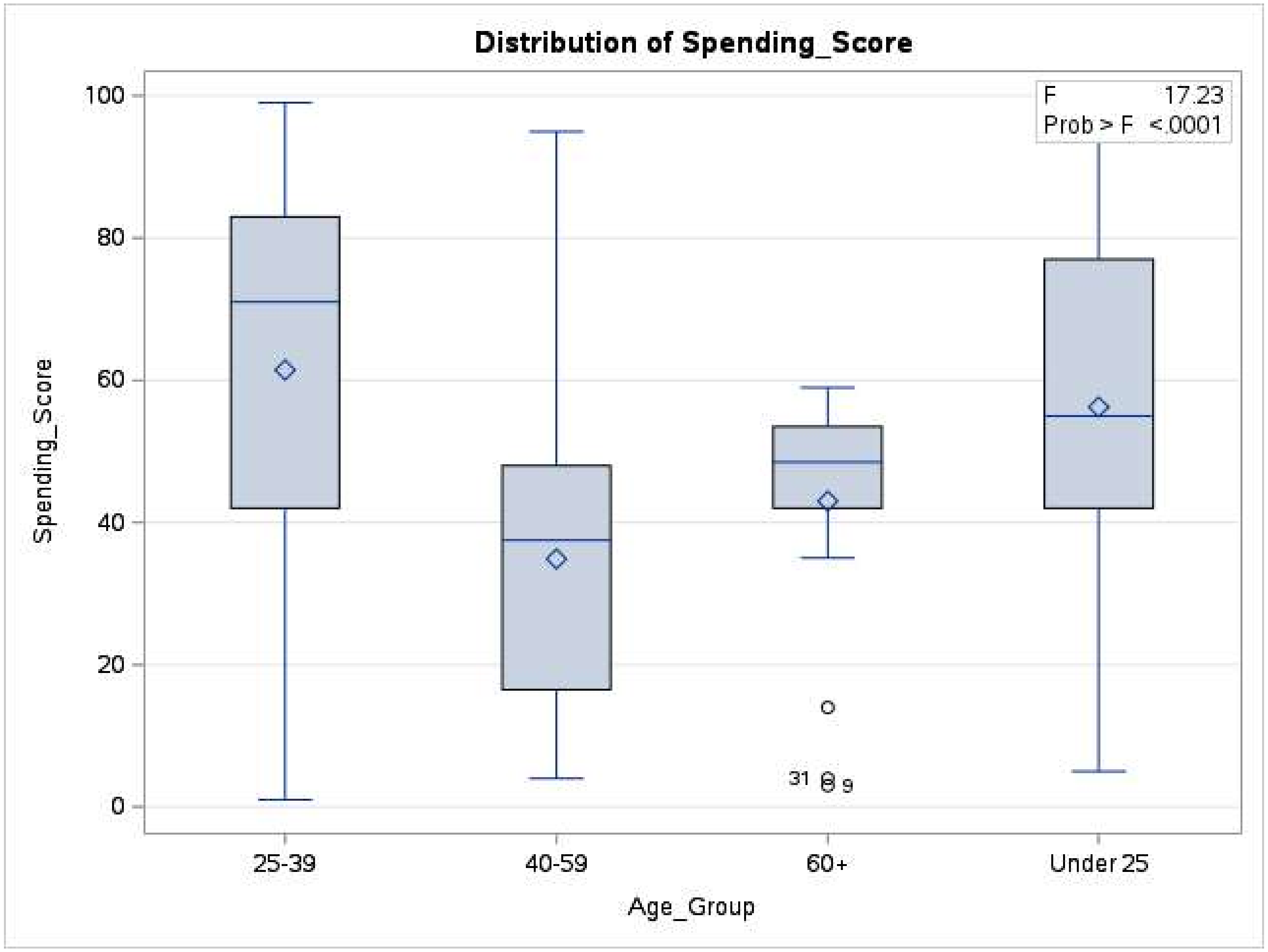
Dependent Variable: Spending_Score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	27691.3577	9230.4526	17.23	<.0001
Error	196	105012.6423	535.7788		
Corrected Total	199	132704.0000			

R-Square	Coeff Var	Root MSE	Spending_Score Mean
0.208670	46.10935	23.14690	50.20000

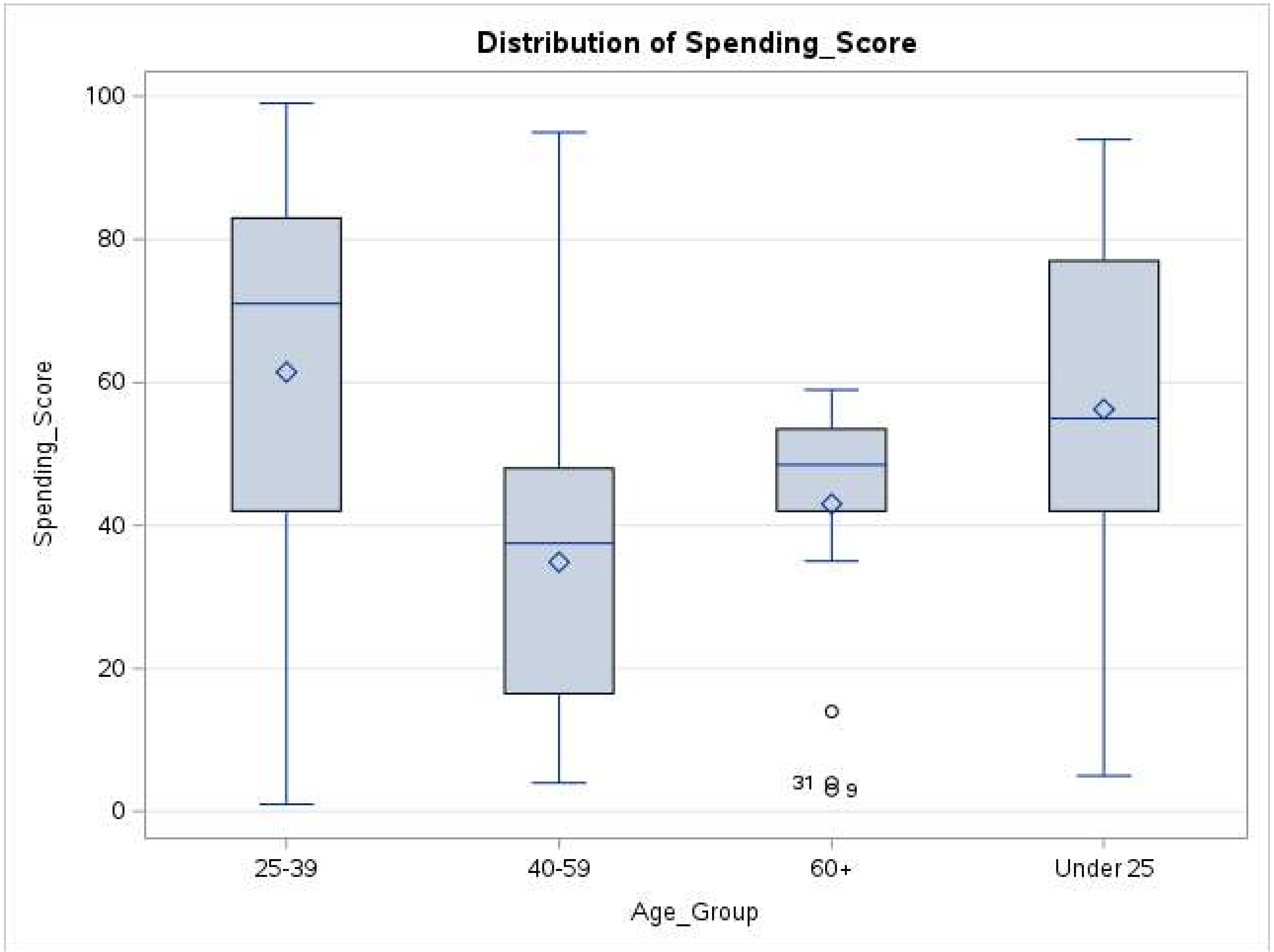
Source	DF	Type I SS	Mean Square	F Value	Pr > F
Age_Group	3	27691.35769	9230.45256	17.23	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Age_Group	3	27691.35769	9230.45256	17.23	<.0001



Tukey HSD Post-Hoc for Age_Group ANOVA

The GLM Procedure



Tukey HSD Post-Hoc for Age_Group ANOVA

The GLM Procedure

Tukey's Studentized Range (HSD) Test for Spending_Score

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	196
Error Mean Square	535.7788
Critical Value of Studentized Range	3.66452

Comparisons significant at the 0.05 level are indicated by ***.				
Age_Group Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
25-39 - Under 25	5.224	-6.908	17.357	
25-39 - 60+	18.481	3.505	33.458	***
25-39 - 40-59	26.622	16.591	36.653	***
Under 25 - 25-39	-5.224	-17.357	6.908	
Under 25 - 60+	13.257	-3.555	30.069	
Under 25 - 40-59	21.398	8.789	34.007	***
60+ - 25-39	-18.481	-33.458	-3.505	***
60+ - Under 25	-13.257	-30.069	3.555	
60+ - 40-59	8.141	-7.224	23.506	
40-59 - 25-39	-26.622	-36.653	-16.591	***
40-59 - Under 25	-21.398	-34.007	-8.789	***
40-59 - 60+	-8.141	-23.506	7.224	